

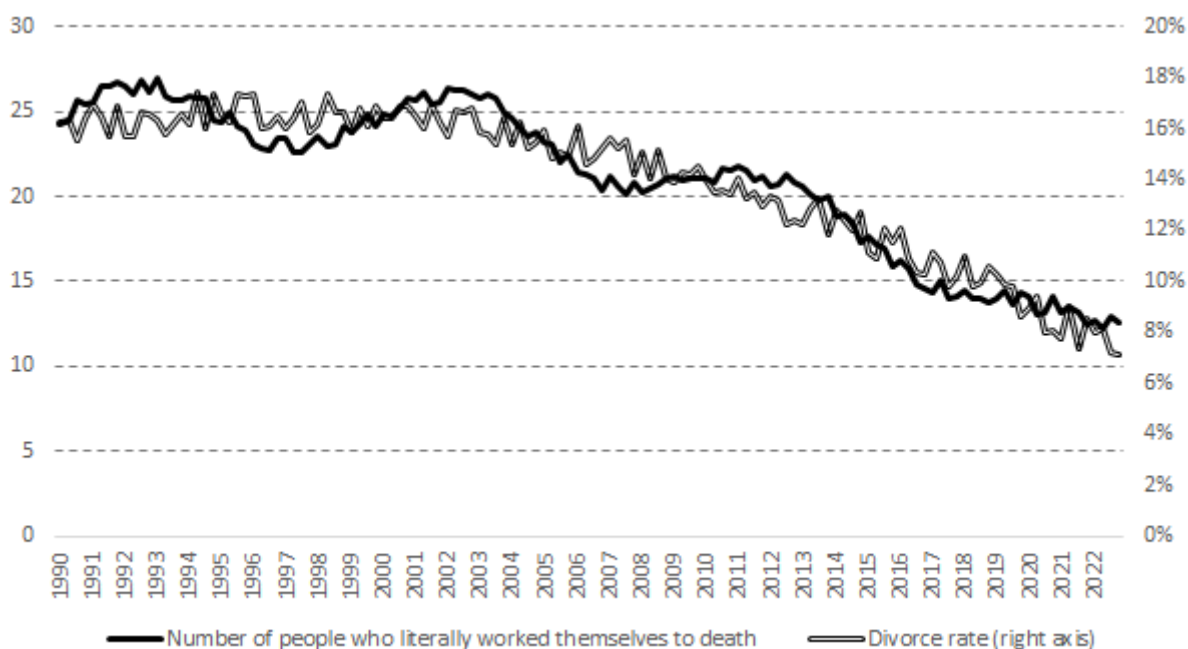
Advanced Econometrics – Exam

June 17, 2024

Exercise 1 (40%)

An economist was interested in a relationship between the divorce rate and the number of people who literally worked themselves to death for a certain Western country. Figure 1 presents these two time series for the period of 1990Q1-2022Q4. The economist figured out to perform a cointegration analysis. To do so she used a data frame `divrate` that consists of two variables `divrate$divorce_rate` and `divrate$work_to_death`.

Figure 1: Number of people who literally worked themselves to death and the divorce rate (right scale)



Use $\alpha = 5\%$ significance level.

- (5%) Was it a good idea to use two scales instead of one scale plot? Explain.
- (5%) Does visual inspection prove the idea to estimate the cointegration model? Explain.
- (10%) Are these two variables cointegrated? Explain.
- (10%) Interpret the estimates of the error-correction mechanism.
- (10%) Is this relationship real or spurious? Explain.

The critical value for the cointegration test should be -2.50.

```
> testdf2(variable = dirate$work_to_death, test.type = "nc",
+         max.augmentations = 4, max.order = 4)
      augmentations      adf      p_adf bgodfrey.statistic bgodfrey.statistic.1
1           0 -1.787547 0.07403626           131      1.456982e+01
2           1 -2.533419 0.01261699           130      1.070850e+00
3           2 -1.244677 0.15406712           129      7.581982e-02
4           3 -0.345037 0.71172303           128      7.326484e-04
5           4 -3.136046 0.01000000           127      1.846717e-03
      bgodfrey.statistic.2 bgodfrey.statistic.3 bgodfrey.statistic.4 p_bg.p.value.1
1      22.44028900      22.98759338      23.0073841 0.0001350605
2      10.35119490      10.38145928      10.4317578 0.3007535801
3       0.22806208       0.68298753       1.1599404 0.7830442955
4       0.04073514       0.04141187       0.2392068 0.9784059131
5       0.07135416       0.07451943       0.1655916 0.9657226983
      p_bg.p.value.2 p_bg.p.value.3 p_bg.p.value.4
1 0.0000134014 0.000040624 0.0001261968
2 0.0056528390 0.015586970 0.0337501041
3 0.8922303000 0.877198000 0.8846491000
4 0.9798384000 0.997786300 0.9933930356
5 0.9649518120 0.999470900 0.9967558751
```

```
> testdf2(variable = xts.diff(dirate$work_to_death), test.type = "nc",
+         max.augmentations = 4, max.order = 4)
      augmentations      adf      p_adf bgodfrey.statistic bgodfrey.statistic.1
1           0 -15.634660 0.01           130      2.267298
2           1 -11.247126 0.01           129      1.395060
3           2 -7.924754 0.01           128      1.409979
4           3 -5.846640 0.01           127      1.081811
5           4 -4.889280 0.01           126      1.486934
      bgodfrey.statistic.2 bgodfrey.statistic.3 bgodfrey.statistic.4 p_bg.p.value.1
1      13.298095      13.458748      13.887250 0.1321301
2       2.813597       3.666395       3.669316 0.2375525
3       3.101025       4.200200       4.206556 0.2350598
4       3.343964       5.487451       6.781993 0.2982927
5       4.276810       6.831667       8.533663 0.2226927
      p_bg.p.value.2 p_bg.p.value.3 p_bg.p.value.4
1 0.001295255 0.003742613 0.007663739
2 0.244926160 0.299813790 0.452604493
3 0.212139193 0.240641874 0.378772623
4 0.187874353 0.139391111 0.147867232
5 0.117842656 0.077461102 0.073873398
```

```
> pp.test <- ur.pp(dirate$div_rate, type = c("Z-tau"), model = c("constant"))
> summary(pp.test)
```

```
#####
# Phillips-Perron Unit Root Test #
#####
Test regression with intercept
```

Value of test-statistic, type: Z-tau is: 0.4607

aux. Z statistics

Z-tau-mu 1.4212

Critical values for Z statistics:

	1pct	5pct	10pct
critical values	-3.500629	-2.892174	-2.582905

```
> kpss.test <- ur.kpss(diff.xts(dirate$div_rate), type = c("mu"))
> summary(kpss.test)
```

```
#####
# KPSS Unit Root Test #
#####
```

Test is of type: mu with 3 lags.

Value of test-statistic is: 0.2267

Critical value for a significance level of:

	10pct	5pct	2.5pct	1pct
critical values	0.347	0.463	0.574	0.739

```
> model <- lm(div_rate~work_to_death, data=dirate)
> summary(model)
```

Call:

lm(formula = div_rate ~ work_to_death, data = dirate)

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.10888	0.54258	-0.201	0.841
work_to_death	1.26277	0.03158	39.991	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.226 on 130 degrees of freedom

Multiple R-squared: 0.9248, Adjusted R-squared: 0.9242

F-statistic: 1599 on 1 and 130 DF, p-value: < 2.2e-16

```
> testdf2(variable = residuals(model), test.type="nc",
+         max.augmentations = 4, max.order = 4)
  augmentations      adf      p_adf bgodfrey.statistic bgodfrey.statistic.1
1             0 -3.388754 0.01000000                94          0.93042284
2             1 -2.649011 0.01000000                93          0.02772583
3             2 -2.776510 0.01000000                92          0.03700397
4             3 -2.285588 0.02315666                91          0.24036069
5             4 -3.004107 0.01000000                90          0.01878585
  bgodfrey.statistic.2 bgodfrey.statistic.3 bgodfrey.statistic.4 p_bg.p.value.1
1          2.7945136          4.4562976          10.6959095          0.3347531
2          0.4497801          2.0040100          7.9592378          0.8677550
3          0.4075956          1.2338363          8.4125294          0.8474570
4          0.3039606          0.3366230          5.6099722          0.6239457
5          0.1223305          0.4198245          0.5794104          0.8909822
  p_bg.p.value.2 p_bg.p.value.3 p_bg.p.value.4
1          0.2472744          0.2162217          0.03020276
2          0.7986040          0.5715748          0.09308283
3          0.8156273          0.7449009          0.07758338
4          0.8590052          0.9529997          0.23023062
5          0.9406678          0.9361174          0.96532395
```

```
> model.ecm <- lm(diff_div_rate~diff_work_to_death + lresid - 1, data = dirate)
> summary(model.ecm)
```

Call:

```
lm(formula = diff.xts(div_rate) ~ diff.xts(work_to_death) + lresid - 1,
    data = dirate)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
diff_work_to_death	1.25168	0.03905	32.056	<2e-16 ***
lresid	-0.14310	0.01379	-10.38	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Exercise 2 (30%)

A different economist was intrigued by what factors affect the number of kids per family. The dependent variable in her analysis was `kids` measured the number of kids (count). As regressors, she decided to use: `ln_income` - the logarithm of income, `age_head` - age of the head of the family minus 21, `tenure` - the number of years the couple has been together, and `cars` - number of vehicles the family owns.

The quality publication table below presents estimates of five models. The columns `pois1` and `pois2` contain the Poisson regressions estimates, `nbreg1` and `nbreg2` contain the negative binomial regression estimates, and finally column `zip` includes the zero-inflated Poisson model.

Use $\alpha = 10\%$ **significance level**. The next page starts with the chi-square distribution critical values.

- (5%) Interpret the parameters of `pois2` model.
- (5%) Do we have an excess of zeros? Explain referring to appropriate statistics or tests.
- (5%) Do we have overdispersion or underdispersion? Explain.
- (5%) Which model should be selected from the quality publication table as the best for modeling the number of kids? Explain.
- (10%) The economist read a textbook on count data models by Cameron and Trivedi (2013), which discusses the possibility of using ordered models for count data. Does the ordered logit presented on page 10 give the same conclusions as count models from the table on page 8?

	pois1 b/se	pois2 b/se	nbreg1 b/se	nbreg2 b/se	zip b/se
kids					
ln_income	0.104*** (0.038)	0.103*** (0.038)	0.104*** (0.038)	0.103*** (0.038)	0.111*** (0.038)
age_head	0.059*** (0.006)	0.059*** (0.006)	0.059*** (0.006)	0.059*** (0.006)	0.059*** (0.006)
tenure	0.016* (0.009)	0.016* (0.009)	0.016* (0.009)	0.016* (0.009)	0.021** (0.009)
cars		0.003 (0.014)		0.003 (0.014)	
_cons	0.113 (0.113)	0.108 (0.116)	0.113 (0.113)	0.108 (0.116)	0.077 (0.113)
inflate					
ln_income					6.458 (4.088)
age_head					0.635 (0.502)
tenure					2.725* (1.647)
cons					-53.610* (32.266)
chi2	102.581	102.616	98.040	98.074	102.242
p-value	0.000	0.000	0.000	0.000	0.000
loglikelihood	-1914.206	-1914.206	-1911.935	-1911.935	-1909.485
N	1000.000	1000.000	1000.000	1000.000	1000.000
AIC	3733.830	3735.796	3735.830	3737.796	3732.728
BIC	3753.461	3760.334	3760.369	3767.242	3771.990

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Upper-tail critical values of the chi-square distribution

degrees of freedom	Probability less than the critical value				
	0.90	0.95	0.975	0.99	0.999
1	2.706	3.841	5.024	6.635	10.828
2	4.605	5.991	7.378	9.210	13.816
3	6.251	7.815	9.348	11.345	16.266
4	7.779	9.488	11.143	13.277	18.467
5	9.236	11.070	12.833	15.086	20.515
6	10.645	12.592	14.449	16.812	22.458

```
> ks.test(kids, "ppois", mean(kids))
One-sample Kolmogorov-Smirnov test
data: kids
D = 0.22509, p-value < 2.2e-16
alternative hypothesis: two-sided
```

```
> zero.test(kids)
Score test for zero inflation
Chi-square = 0.19622
df = 1
pvalue: 0.65779
```

```
> lrtest(pois1, nbreg1)
Likelihood ratio test

Model 1: kids ~ ln_income + age_head + tenure
Model 2: kids ~ ln_income + age_head + tenure
#Df LogLik Df Chisq Pr(>Chisq)
1 4 -1862.9
2 5 -1862.9 1 0.0011 0.9733
```


The output below presents R output of the ordered logit model. In this model, there are 11 thresholds, because the dependent variable kids takes on values in the set $\text{kids} \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$.

```
> ologit = ologit.reg(kids~ln_income+age_head+tenure)
> summary(ologit)
Ordered Logit Regression
Log-Likelihood: -1857.757
No. Iterations: 6
McFadden's R2: 0.6150369
AIC: 3743.515
```

	Estimate	Std. error	t value	Pr(> t)	
ln_income	0.287769	0.112623	2.5552	0.01061	*
age_head	0.177826	0.018777	9.4704	< 2e-16	***
tenure	0.042645	0.025120	1.6977	0.08957	.

```
----- Threshold Parameters -----
```

	Estimate	Std. error	t value	Pr(> t)	
Threshold (0->1)	-0.18627	0.33779	-0.5514	0.5813	
Threshold (1->2)	1.44368	0.32833	4.3970	1.098e-05	***
Threshold (2->3)	2.53252	0.33376	7.5878	3.254e-14	***
Threshold (3->4)	3.46233	0.34169	10.1331	< 2.2e-16	***
Threshold (4->5)	4.42845	0.35258	12.5600	< 2.2e-16	***
Threshold (5->6)	5.51513	0.37130	14.8535	< 2.2e-16	***
Threshold (6->7)	6.16400	0.39197	15.7256	< 2.2e-16	***
Threshold (7->8)	7.06470	0.44675	15.8134	< 2.2e-16	***
Threshold (8->9)	7.85166	0.53853	14.5798	< 2.2e-16	***
Threshold (9->10)	8.95711	0.78968	11.3427	< 2.2e-16	***
Threshold (10->11)	9.65121	1.06000	9.1049	< 2.2e-16	***

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```


Exercise 3 (30%)

Lewkowicz¹ et al. (2024) considered factors affecting constitutional compliance.

”Although compliance with constitutions is undoubtedly one of the most essential topics in the current public debate, there are still a lot of questions about its determinants that have to be addressed. What is the impact of the populist and anti-pluralistic character of ruling political parties on constitutional compliance of the government and does it depend on how long such parties have been in power? In our study, based on data for 147 countries during the period between 1990 and 2019, we aim to investigate the relationships between populism, anti-pluralism, and constitutional compliance, with a special focus on the years of the populist or anti-pluralistic ruling parties in office.”

In this exercise, we will focus on two (out of four) hypotheses from the article:

”Hypothesis 2: The length of the term in office of the government using populist rhetoric is not a significant predictor of its constitutional compliance.

Hypothesis 3: The more anti-pluralist political beliefs are presented by the party before the elections, the lower its constitutional compliance is as a ruling party after the elections.”

”The main dependent variable is **cc_total**, and it comes from the Comparative Constitutional Compliance Database (CCCD). It is designed to capture, inter alia, how often the government violates the constitution, and whether its actions are subject to legal proceedings. [...] The primary independent variable, the Populism index (Populism), describes the extent to which populist rhetoric is used by representatives of the ruling party. [...] The V-Dem dataset served as a source of variables regarding the political polarization (polarization), lower chamber election seat share won by the largest party (Seats_share), participation of civil society in the ruling process (Civil_soc), international autonomy of the state (Internat_auton), importance of the person of the leader (Legit_leader), number of seats won in the lowest chamber by the largest party (Seat share), and estimate of the GDP per capita (Log_GDP_pc). V-Party was used to obtain measures of the ruling party’s anti-pluralism index (Anti-pluralism), and the position on the left-right economic scale (Left-right). The variable years in office (years) describing the duration of the given party’s rule was computed and added to the dataset. For any given year, the variable represents the number of years the governing party remains in office uninterruptedly. Supplementary socio-economic measures regarding life expectancy, the Human Capital Index, alternative GDP per capita measures, and unemployment and inflation rates were obtained from the World Bank and the International Monetary Fund databases. The first two variables were sourced with the sole purpose of instrumenting the GDP per capita variable. We also account for the type of regime by referring to the presence of electoral democracy (a dummy variable Democracy; 1=Democracy; 0=Nondemocracy).”

Use $\alpha = 5\%$ significance level. Please mind the star levels used in the quality publication table.

- a) (5%) Interpret parameters of model1.
- b) (5%) What are the conclusions from the Hausman test results? Explain
- c) (10%) Which model should be selected as superior out of those presented? Explain.
- d) (5%) How to verify Hypothesis 2? Explain.
- e) (5%) How to verify Hypothesis 3? Explain.

¹Lewkowicz, J., Woźniak, R., Lewczuk, A., & Marcol, M. (2024). The longer the worse? The case of populism, anti-pluralism, and constitutional compliance, *Democratization*, 31:1, 132-156, DOI: 10.1080/13510347.2023.2258341

	model1	model2	model3	model4
Democracy	0.3322*** (0.0775)	0.1858** (0.0727)	0.6422*** (0.0703)	0.2490*** (0.0655)
Populism	0.0673 (0.0768)	0.2022** (0.0795)	0.0342 (0.0777)	0.1941** (0.0758)
years	-0.0041 (0.0029)	-0.0012 (0.0028)	-0.0068** (0.0028)	-0.0006 (0.0025)
Anti-pluralism		-0.3804*** (0.0865)		-0.3949*** (0.0828)
ln_GDP_pc		0.1126* (0.0636)		0.0918** (0.0377)
Left_right		0.0240* (0.0127)		0.0189 (0.0124)
Polarization		-0.1118*** (0.0256)		-0.0987*** (0.0223)
Seats_share		0.0003 (0.0012)		-0.0004 (0.0011)
Civil_soc		1.3415*** (0.2401)		1.5722*** (0.1974)
Internat_auton		-0.0524 (0.0579)		-0.0389 (0.0498)
Legit_leader		-0.1456*** (0.0301)		-0.1342*** (0.0258)
Inflation		-0.0032 (0.0020)		-0.0031 (0.0019)
Constant	0.3602*** (0.0641)	-0.6514** (0.2661)	0.0893 (0.0857)	-0.7857*** (0.2016)
N	669.0000	608.0000	669.0000	608.0000
N_g	155.0000	147.0000	155.0000	147.0000
F	7.4660	21.6044		
chi2			97.3474	515.2505
model	fe	fe	re	re
R2_b	0.4714	0.6273	0.4821	0.6428
R2_w	0.0420	0.3660	0.0411	0.3630
R2_o	0.4047	0.5746	0.4145	0.5872

* p<0.10, ** p<0.05, *** p<0.010

```
> phptest(model1, model3)
```

Hausman Test

```
data: prod ~ area + labor + fert  
chisq = 60.22, df = 3, p-value = 2.2e-16  
alternative hypothesis: one model is inconsistent
```

```
> phptest(model2, model4)
```

Hausman Test

```
data: prod ~ area + labor + fert  
chisq = 23.34, df = 12, p-value = 0.0250  
alternative hypothesis: one model is inconsistent
```